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15 April 1960

MILITARY SPECIFICATION

CONTROL, PUBLIC ADDRESS SET C-1614/AIC-13

1. SCOPE.

1.1 This specification covers one type of control, designated Control, Public Address Set C-1614/AIC-13, which is one item of Public Address Set AN/AIC-13.

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-C-5015	Connector, Electric AN Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-C-6781	Control Panel; Aircraft Equipment, Rack and Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-S-19500	Semiconductor Devices, General Specification For

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electronic Power, Aircraft, Characteristics and Utilization
MIL-STD-781	Reliability Tests Exponential Distribution

FSC 5831

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DRAWINGS

U. S. Air Force

60C12546	Wiring Diagram-Headset Power Amplifier, Schematic
56C12707	Control Public Address Set C1614/AIC-13 Outline Dimensions Of
56C12708	Wiring Diagram - C1614/AIC-13 Public Address Set Control, Schematic
56D12709	Wiring Diagram - C1614/AIC-13 Public Address Set Control Test Circuit, Schematic
56J12718	Installation Information, Wiring Diagram, Public Address Set AN/AIC-13
68C34500	Panel, Plastic Lighting, C1614/AIC-13
67B45901	Knob, Skirt Assembly Control Panel
67B45902	Dial Skirt, Knob Control Panel
67B45950	Knob Assembly - Control Panel

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General Specification. The requirements of MIL-E-5400 for Class 2 equipment apply as requirements of this specification with the exceptions and additions called out herein. When the two conflict, this specification shall govern.

3.2.1 Substitution of Material. If a manufacturer, either prior to or during production, desires to substitute another material or process for one previously approved or specified, he shall submit to the procuring activity a statement to that effect. The statement shall describe the proposed substitution and the reason therefor, together with evidence to show that the substitution is satisfactory. At the discretion of the procuring activity, test samples or additional data may be required to prove the adequacy of the proposed substitution.

3.2.2 Electrical Connector. The external electrical connector shall meet the requirements of AF Drawing 56C12708 and MIL-C-5015.

3.2.3 Volume Control. The volume control shall be a step type, mechanically linked to a d-c motor for remote operation and shall have provision for manual operation at the control panel. The attenuation shall be 0 decibels

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in the maximum clockwise position and 33 decibels plus or minus 2 decibels in the maximum counterclockwise position. The attenuation change shall be essentially linear in 10 steps of approximately 3.3 decibels each. The total resistance of volume control shall be 500 ohms plus or minus 10 percent. Mechanical and electrical stops shall be provided at the extreme position of the volume control. The volume control knob shall be positioned on its shaft in such a manner that the straight-line reference mark on the knob shall align with the "12 o'clock" position at the center of its range of rotation. The torque required to turn the shaft of the volume control shall be between 25 and 35 ounce inches.

3.2.4 Console Control Panel. The control panel shall be in accordance with the requirements of MIL-C-6781 for Type 1 control panel except as specified herein.

3.2.4.1 Panel Layout. Layout of component parts and controls on the front surface of the control panel shall be in accordance with Drawings 56C12707, 68C34500 and MIL-P-7788 Class 1-R Type 3.

3.2.5 Circuit Label. The circuit label shall be similar to the schematic diagram shown on Drawings 56C12708, and 60C12546 and shall include word identification of the terminals. The complete nomenclature of Control, Public Address Set C-1614/AIC-13 shall be used to identify the circuit label, which shall be firmly secured to the inside surface of the top cover. A sketch of the proposed circuit label shall be submitted to the procuring activity for approval prior to fabrication of the production circuit labels.

3.2.6 Service Conditions (Electrical). The equipment shall be designed to operate from a power source with electrical characteristics conforming to the d-c power requirements of MIL-STD-704 and category B, Table II steady state limits.

3.2.7 Transistors. The transistors shall meet the requirements of MIL-S-19500.

3.2.8 Wiring. The control panel shall be wired in accordance with Drawings 56C12708 and 60C12546.

3.2.8.1 Crosstalk. Cabling shall be such that interference resulting from crosstalk is at least 50 db down when tested as specified herein.

3.2.9 Electromagnetic Interference. Electromagnetic interference control testing shall be in accordance with the applicable requirements of MIL-STD-461, Table I, for Class IC equipment.

3.3 Reliability. The reliability requirements of MIL-STD-781 shall be met as specified herein. The contract specified mean-time-between-failure (MTBF) shall

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not be less than 6000 hours.

3.4 Dimensions. The overall dimensions of the assembled control shall be in accordance with Drawings 56C12707 and 68C34500.

3.4.1 Weight. The weight of the C-1614/AIC-13 control shall not exceed 2.75 pounds.

3.5 Performance.

3.5.1 Amplifier. When tested as specified herein, the amplifier shall be capable of the following performance:

3.5.1.1 Input-Output Characteristics. The input-output characteristic at 1000 H_z shall be within the limits of Table I.

TABLE I

Input to 150 ohms Volts	Nominal Output Volts	Minimum Output Volts	Maximum Output Volts
0.5	1.5	1.1	2.2
1.0	3.0	2.2	4.5
2.0	6.0	4.5	9.0
3.0	9.0	6.2	12.5
4.0	12.0	9.0	16.0

3.5.1.2 Frequency-Response Characteristics. With a constant input of 2.0 volts rms to the input of 150 ohms, the output of the amplifier relative to 1000 H_z shall be within the limits of Table II.

TABLE II

Frequency Hz	Output (db relative to 1000 Hz)
300	Plus 1, minus 4.5
400	Plus 1.0, minus 3.5
1000	0 (reference)
2000	Plus 1.0, minus 2.5
4000	Plus 1.0, minus 3.5
6000	Plus 0, minus 5.0
10000	Minus 1.0, minus 9.0

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3.5.1.3 Distortion. The total harmonic distortion at 300 Hz shall not exceed the values specified in Table III.

TABLE III

Output Volts	Maximum Distortion (Percent)
1.5	3.0
3.0	4.0
6.0	6.0
11.0	12.0

3.5.1.4 Power Output. The maximum power output of the amplifier at 1000 Hz shall not be less than 1.0 watt.

3.5.1.5 Internal Noise Level. The noise output of the amplifier with no ripple voltage applied to the 27.5 volt d-c line shall not exceed 0.005 volt.

3.5.1.6 Microphonic Noise. When subject to the specified vibration cycling, the microphonic noise output of the amplifier shall not exceed 0.01 volt.

3.5.2 Operation. A typical interconnection of Control, Public Address Set C-1614/AIC-13 with the remaining components of Public Address Set AN/AIC-13 is shown in Drawing 56J12718. The control shall provide satisfactory operation and shall not adversely affect the performance of any other component of Public Address Set AN/AIC-13 when interconnected with, and operated as a part of that equipment.

3.6 Government Loaned Property. When specified in the contract or purchase Order, and if available, the following items will be furnished by the Government on loan to the contractor upon his request:

Item Number	Item Description	Quantity
1	Amplifier, Audio Frequency AM-944/AIC-13	1
2	Control, Public Address Set C-1614/AIC-13	1
3	Loudspeaker, Magnetic LS-211/AIC-13	1

4. QUALITY ASSURANCE PROVISIONS.

4.1 Classification of Tests. The inspection and testing of the C-1614/AIC-13 Control shall be classified as follows:

a. Preproduction tests . . . . . See 4.3

b. Acceptance tests . . . . . See 4.4

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4.2 Test Conditions. Unless otherwise specified, the amplifier shall be tested under the following conditions:

a. Temperature: Room ambient, plus 15°C to plus 35°C

b. Pressure: Normal atmospheric

c. Humidity: Room ambient, up to 90 percent relative humidity

d. Primary voltage supply: 27.5 volts d-c ± 1.0 volt

4.3 Preproduction Tests. Preproduction tests shall be performed by the contractor in accordance with the following:

4.3.1 Preproduction Test Samples.

4.3.1.1 Contractor Test Samples. When specified in the contract a minimum of 10 each C-1614/AIC-13 controls shall be subjected by the contractor to the preproduction tests specified herein. Seven (7) each of these samples shall be used for reliability testing only.

4.3.1.2 Government Test Samples. When specified in the contract, the 3 each controls not used above in reliability testing shall be submitted to the procuring activity. The samples shall be accompanied by adequate identifying information including the Government order number involved ("File Number" in contract), if any, name and address of prime contractor, and name and address of manufacturer. The samples and data will be used by the procuring activity for:

a. Review of the mechanical design.

b. Performance of any tests, specified herein, which may be deemed necessary after a review of the contractor's test report.

c. Government reference purposes.

4.3.2 Test Schedule. The equipment shall be subjected to the tests listed in Table IV in the quantities shown. The order of conducting the tests need not be in order listed in the table.

TABLE IV

Inspection or Test	Paragraph	Pre-production	Individual	Sampling
Examination of Product	4.5.1	3	All	All
Electrical Tests	4.5.2			

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Input-output characteristic	3.5.1.1	3		All
Input-output characteristic test	4.5.2.1		All	
Frequency response characteristic	3.5.1.2	3		All
Frequency response Characteristic test	4.5.2.2		All	
Power output	3.5.1.4	3		All
Distortion	3.5.1.3	3	All	All
Distortion Test	4.5.2.3		All	
Internal noise level	3.5.1.5	3		All
Microphonic noise	3.5.1.6	3		All
Circuitry test	4.5.2.4	3	All	All
Crosstalk Characteristic	3.2.8.1	3		All
Crosstalk test	4.5.2.5	3		All
Operation test	4.5.3	3		All
Electromagnetic Interference	3.2.9	3		
Reliability	3.3			
Reliability Test	4.5.5	7		
Environmental Tests	4.5.4			
Temperature-Altitude		2		All
Vibration		2		
Shock		2		
Humidity		2		All
Salt Spray		2		
Temperature Shock		2		
Sand and Dust		2		
Fungus		2		All

NOTE: Where a Paragraph in Section 3 is referenced above, the test procedure shall consist of determining compliance with the requirements stated in that paragraph.

4.4 Acceptance Test. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.4.1 Individual Tests. Each Control shall be subjected to the individual tests listed in Table IV.

4.4.2 Sampling Tests. Unless otherwise specified sampling tests shall be conducted on at least one out of every 100 controls, or fractional part thereof, on order. The samples shall be selected at random from quantities of controls which have passed the individual tests. The samples shall be subjected to the sampling tests listed in Table IV in the quantities shown.

4.4.2.1 Acceptance Criteria. When one or more of the selected controls fails to meet the specified requirements, acceptance of all items shall be

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withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5 Test Procedures.

4.5.1 Examination of Product. The control shall be examined to determine compliance with the requirements of this specification for materials, workmanship, dimensions, and markings.

4.5.2 Electrical Tests. The controls shall be electrically tested to determine compliance with this specification using the test circuit shown on Drawing No. 56C12709 or a similar test circuit, which has been approved by the procuring activity.

4.5.2.1 Input-Output Characteristic Test. The output of the control shall be measured at 1000 Hz with an input of 2.0 volts to one of the mixing input terminals.

4.5.2.2 Frequency Response Test. The frequency response shall be measured at 300 and 6000 Hz with an input of 2.0 volts to one of the mixing input terminals.

4.5.2.3 Distortion Test. The distortion in the output of the control shall be measured when the amplifier is developing 11.0 volts across the load resistor at 300 Hz.

4.5.2.4 Circuitry Test. The control shall be tested for continuity of wiring and correctness of wiring and volume control action.

4.5.2.5 Crosstalk Test. Crosstalk between mixer input lines shall be determined by feeding a 2 volt 1000 Hz signal to a mixer input line and measuring the crosstalk voltage across each of the remaining mixer input lines.

4.5.3 Operation. The control shall be connected into a representative AN/AIC-13 system and shall be tested operationally to determine the extent of compliance with the requirements of 3.5.2.

4.5.4 Environmental Tests. The control shall be tested for conformance with the following tests of military specification MIL-E-5272:

Temperature-Altitude	Procedure I
Vibration	Procedure XII
Shock	Procedure V
Humidity	Procedure I
Salt Spray	Procedure I

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Sand and Dust	Procedure I
Temperature Shock	Procedure I
Fungus	Procedure I

4.5.4.1 Criteria For Failure. The inability of an amplifier immediately after each environmental test to meet, within 15 percent, the requirements of the individual tests of Table IV shall constitute a failure.

4.5.5 Reliability Preproduction Test.

4.5.5.1 Reliability Test. Reliability test as required by Paragraph 4.3.2 shall be conducted on Test Level F from MIL-STD-781.

4.5.5.2 Temperature Cycle. Temperature cycling shall be Standard Method of temperature cycling as specified in MIL-STD-781. The control shall be maintained at the upper test level temperature for two (2) hours.

4.5.5.3 Specific Duty Cycle. Specific duty cycle shall be in accordance with MIL-STD-781.

4.5.5.4 Voltage Cycling. Voltage cycling shall be in accordance with MIL-STD-781.

4.5.5.5 Equipment Operating Parameters. The operating parameters shall be as specified in the individual tests of Table IV. Failure to meet the minimum requirements specified shall constitute a failure. Measurement of operating parameters shall be in accordance with MIL-STD-781.

4.5.5.6 Test Plan. Test Plan V in accordance with MIL-STD-781, Table 2 with $\theta_0 = 6000$ and $\theta_1 = 2000$.

4.5.5.7 Test Option. The contractor has the option of running the reliability tests simultaneously with the other required preproduction tests with the stipulation that if any change is found needed by the other tests the reliability tests must be restarted with zero time. If a failure occurs it will be analyzed, repaired and replaced on test.

5. PREPARATION FOR DELIVERY.

5.1 General. Preparation for delivery shall be in accordance with provisions specified in contract.

5.2 Shipment Marking. Shipment marking nomenclature shall be Control, Public

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Address Set C-1614/AIC-13.

6. NOTES.

6.1 Intended Use. Control, Public Address Set C-1614/AIC-13 is intended for use with Public Address Set AN/AIC-13 to select and monitor five input signals, to control the volume of the output signal, and to select the loudspeaker or group of loudspeakers connected to the output of the power amplifier.

6.2 Ordering Data. Invitations for bids, contract, and purchase order will specify that ten (10) each Control, Public Address Set C-1614/AIC-13 shall be required as preproduction samples to be subjected to the preproduction tests to determine compliance with the requirements of this document, and that three (3) each Control, Public Address Set C-1614/AIC-13 not used in reliability testing shall be submitted to the procuring activity. The invitation for bids and the contract shall specify the point of inspection for these tests.

6.3 Changes From Previous Issue. Astericks are not used in this revision to identify changes with respect to the previous issue due to the extentiveness of the changes.

Custodian:

Air Force - 84

Preparing Activity:

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Project Number:

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

FORM 1426 APR 63 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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